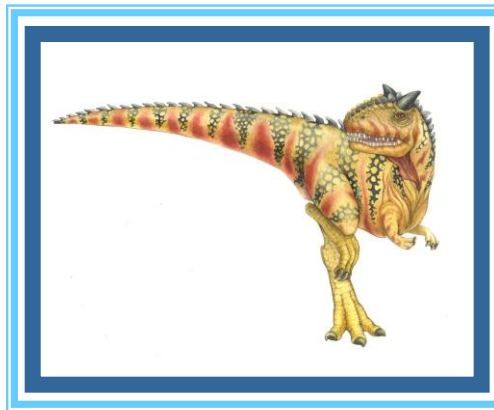


Synchronization





Outline

- Background
- The Critical-Section Problem
- Peterson's Solution
- Hardware Support for Synchronization
- Mutex Locks
- Semaphores
- Monitors
- Liveness
- Evaluation





Objectives

- Describe the critical-section problem and illustrate a race condition
- Illustrate hardware solutions to the critical-section problem using memory barriers, compare-and-swap operations, and atomic variables
- Demonstrate how mutex locks, semaphores, monitors, and condition variables can be used to solve the critical section problem
- Evaluate tools that solve the critical-section problem in low-, Moderate-, and high-contention scenarios





Background

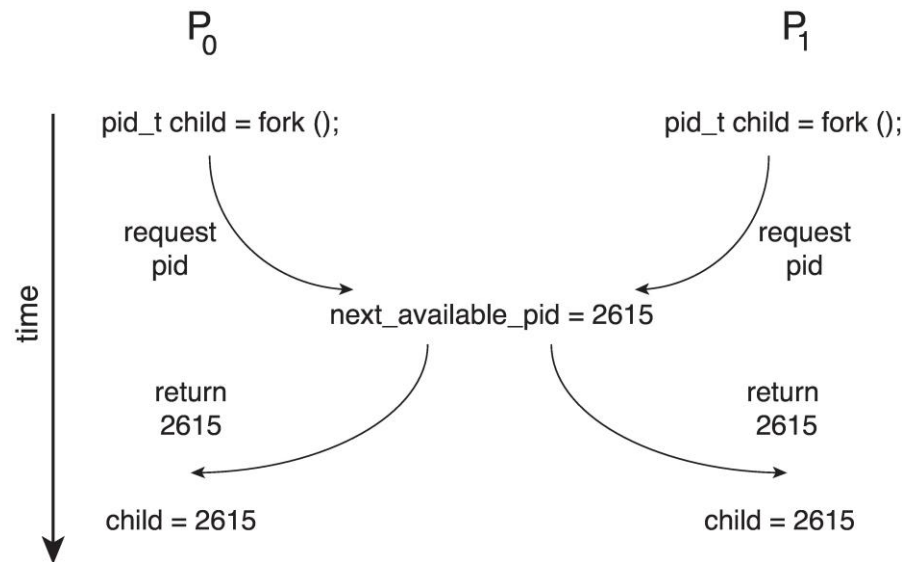
- Processes can execute concurrently
 - May be interrupted at any time, partially completing execution
- Concurrent access to shared data may result in data inconsistency
- Maintaining data consistency requires mechanisms to ensure the orderly execution of cooperating processes
- We illustrated in chapter 4 the problem when we considered the Bounded Buffer problem with use of a counter that is updated concurrently by the producer and consumer,. Which lead to race condition.





Race Condition

- Processes P_0 and P_1 are creating child processes using the `fork()` system call
- Race condition on kernel variable `next_available_pid` which represents the next available process identifier (pid)



- Unless there is a mechanism to prevent P_0 and P_1 from accessing the variable `next_available_pid` the same pid could be assigned to two different processes!





Critical Section Problem

- Consider system of n processes $\{p_0, p_1, \dots p_{n-1}\}$
- Each process has **critical section** segment of code
 - Process may be changing common variables, updating table, writing file, etc.
 - When one process in critical section, no other may be in its critical section
- **Critical section problem** is to design protocol to solve this
- Each process must ask permission to enter critical section in **entry section**, may follow critical section with **exit section**, then **remainder section**





Critical Section

- General structure of process P_i

```
do {  
    entry section  
    critical section  
    exit section  
    remainder section  
} while (true);
```





Critical-Section Problem (Cont.)

Requirements for solution to critical-section problem

1. **Mutual Exclusion** - If process P_i is executing in its critical section, then no other processes can be executing in their critical sections
2. **Progress** - If no process is executing in its critical section and there exist some processes that wish to enter their critical section, then the selection of the process that will enter the critical section next cannot be postponed indefinitely
3. **Bounded Waiting** - A bound must exist on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted
 - Assume that each process executes at a nonzero speed
 - No assumption concerning **relative speed** of the n processes





Interrupt-based Solution

- Entry section: disable interrupts
- Exit section: enable interrupts
- Will this solve the problem?
 - What if the critical section is code that runs for an hour?
 - Can some processes starve – never enter their critical section.
 - What if there are two CPUs?





Software Solution 1

- Two process solution
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- The two processes share one variable:
 - **int turn;**
- The variable **turn** indicates whose turn it is to enter the critical section
- Initially **turn = 1**





Algorithm for Process P_i

```
while (true) {  
    turn = i;  
    while (turn == j)  
        ;  
  
    /* critical section */  
    turn = j;  
  
    /* remainder section */  
}
```





Correctness of the Software Solution

- Mutual exclusion is preserved

P_i enters critical section if and only if:

turn = i

and **turn** cannot be both 0 and 1 at the same time

- What about the Progress requirement?
 - If Process 1 wants to enter the critical section and Process 2 is not interested in entering the critical section, can Process 1 enter?
- What about the Bounded-waiting requirement?





Software Solution -- Peterson's Algorithm

- Two process solution
- Assume that the **load** and **store** machine-language instructions are atomic; that is, cannot be interrupted
- The two processes share two variables:
 - **int turn;**
 - **boolean flag[2]**
- The variable **turn** indicates whose turn it is to enter the critical section
- The **flag** array is used to indicate if a process is ready to enter the critical section.
 - **flag[i] = true** implies that process P_i is ready!





Algorithm for Process P_i

```
while (true){  
  
    flag[i] = true;  
    turn = j;  
    while (flag[j] && turn == j)  
        ;  
  
    /* critical section */  
  
    flag[i] = false;  
  
    /* remainder section */  
  
}
```





Correctness of Peterson's Solution

- Provable that the three CS requirement are met:
 1. Mutual exclusion is preserved
 P_i enters CS only if:
either **flag[j] = false** or **turn = i**
 2. Progress requirement is satisfied
 3. Bounded-waiting requirement is met





Peterson's Solution

- Solution works for 2 process.
- What about modifying it to handle 10 processes?
- Solution requires **busy waiting**
 - Processes waste CPU cycles to ask if they can enter the critical section

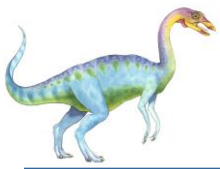




Peterson's Solution and Modern Architecture

- Although useful for demonstrating an algorithm, Peterson's Solution is not guaranteed to work on modern architectures.
 - To improve performance, processors and/or compilers may reorder operations that have no dependencies
- Understanding why it will not work is useful for better understanding race conditions.
- For single-threaded this is OK as the result will always be the same.
- For multithreaded the reordering may produce inconsistent or unexpected results!



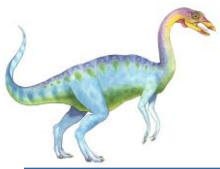


Modern Architecture Example

- Two threads share the data:
`boolean flag = false;`
`int x = 0;`
- Thread 1 performs
`while (!flag)`
`;`
`print x`
- Thread 2 performs
`x = 100;`
`flag = true`
- What is the expected output?

`100`





Modern Architecture Example (Cont.)

- However, since the variables `flag` and `x` are independent of each other, the instructions:

```
flag = true;  
x = 100;
```

for Thread 2 may be reordered

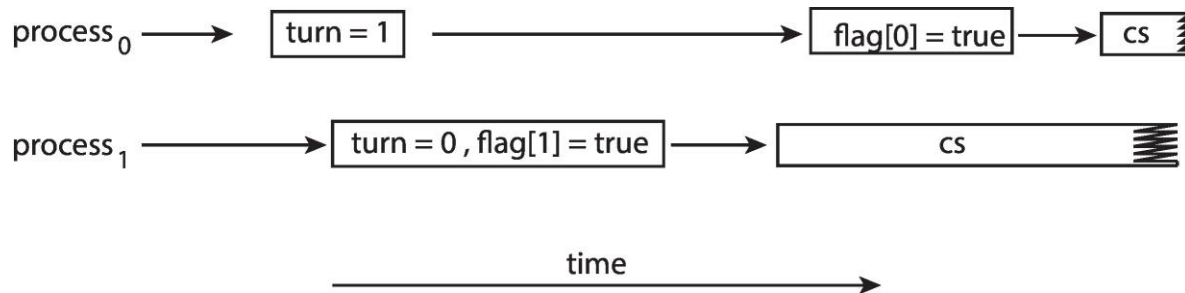
- If this occurs, the output may be
0





Peterson's Solution Revisited

- The effects of instruction reordering in Peterson's Solution



- This allows both processes to be in their critical section at the same time!
- To ensure that Peterson's solution will work correctly on modern computer architecture we must use **Memory Barrier**.





Synchronization Hardware

- Many systems provide hardware support for implementing the critical section code.
- Uniprocessors – could disable interrupts
 - Currently running code would execute without preemption
 - ▶ Is this practical?
 - Generally, too inefficient on multiprocessor systems
 - ▶ Operating systems using this not broadly scalable
- We will look at three forms of hardware support:
 - Memory Barriers
 - Hardware instructions
 - Atomic Variables





Memory Barrier Instructions

- A **memory barrier** instruction is used to ensure that all loads and stores instructions are completed before any subsequent load or store operations are performed.
- Therefore, even if instructions were reordered, the memory barrier ensures that the store operations are completed in memory and visible to other processors before future load or store operations are performed.





Memory Barrier Example

- Returning to the example of slides 6.17 - 6.18
- We could add a memory barrier to the following instructions to ensure Thread 1 outputs 100:
- Thread 1 now performs

```
while (!flag)
    memory_barrier();
print x
```
- Thread 2 now performs

```
x = 100;
memory_barrier();
flag = true
```
- For Thread 1 we are guaranteed that the value of `flag` is loaded before the value of `x`.
- For Thread 2 we ensure that the assignment to `x` occurs before the assignment `flag`.





Hardware Instructions

- Special hardware instructions that allow us to either *test-and-modify* the content of a word, or to *swap* the contents of two words atomically (uninterruptedly).
 - **Test-and-Set** instruction
 - **Compare-and-Swap** instruction





The test_and_set Instruction

■ Definition

```
boolean test_and_set (boolean *target)
{
    boolean rv = *target;
    *target = true;
    return rv;
}
```

■ Properties

- Executed atomically
- Returns the original value of passed parameter
- Set the new value of passed parameter to **true**





Solution Using test_and_set()

- Shared Boolean variable **lock**, initialized to **false**
- Solution:

```
do {  
    while (test_and_set(&lock))  
        ; /* do nothing */  
  
    /* critical section */  
  
    lock = false;  
    /* remainder section */  
} while (true);
```

- Based on busy waiting
- Does it solve the critical-section problem?





The compare_and_swap Instruction

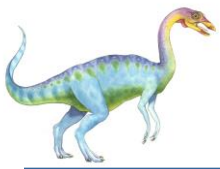
■ Definition

```
int compare_and_swap(int *value, int expected, int new_value)
{
    int temp = *value;
    if (*value == expected)
        *value = new_value;
    return temp;
}
```

■ Properties

- Executed atomically
- Returns the original value of passed parameter **value**
- Set the variable **value** the value of the passed parameter **new_value** but only if ***value == expected** is true. That is, the swap takes place only under this condition.





Solution using compare_and_swap

- Shared integer **lock** initialized to 0;
- Solution:

```
while (true){  
    while (compare_and_swap(&lock, 0, 1) != 0)  
        ; /* do nothing */  
  
    /* critical section */  
  
    lock = 0;  
  
    /* remainder section */  
}
```

- Based on busy waiting
- Does it solve the critical-section problem?





Bounded-waiting with compare-and-swap

```
while (true) {
    waiting[i] = true;
    key = 1;
    while (waiting[i] && key == 1)
        key = compare_and_swap(&lock, 0, 1);
    waiting[i] = false;
    /* critical section */
    j = (i + 1) % n;
    while ((j != i) && !waiting[j])
        j = (j + 1) % n;
    if (j == i)
        lock = 0;
    else
        waiting[j] = false;
    /* remainder section */
}
```





Atomic Variables

- Typically, instructions such as compare-and-swap are used as building blocks for other synchronization tools.
- One tool is an **atomic variable** that provides *atomic* (uninterruptible) updates on basic data types such as integers and Booleans.
- For example:
 - Let **sequence** be an atomic variable
 - Let **increment()** be operation on the atomic variable **sequence**
 - The Command:
increment(&sequence) ;
ensures **sequence** is incremented without interruption:





Atomic Variables

- The `increment()` function can be implemented as follows:

```
void increment(atomic_int *v)
{
    int temp;
    do {
        temp = *v;
    }
    while (temp != (compare_and_swap(v, temp, temp+1)) );
}
```

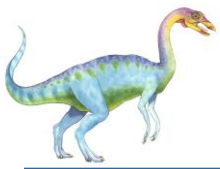




Mutex Locks

- Previous solutions are complicated and generally inaccessible to application programmers
- OS designers build software tools to solve critical section problem
- Simplest is mutex lock
 - Boolean variable indicating if lock is available or not
- Two operations:
 - **acquire()** a lock
 - **release()** a lock
- The **acquire()** and **release()** are executed atomically
 - Usually implemented via hardware atomic instructions such as compare-and-swap.
- But this solution requires **busy waiting**
 - This lock therefore called a **spinlock**





Solution to CS Problem Using Mutex Locks

```
while (true) {  
    acquire lock  
    critical section  
    release lock  
    remainder section  
}
```





Semaphore

- A semaphore is hardware or a software variable whose value indicates the status of a common resource. Its purpose is to lock the resource being used.
- A process which needs the resource will check the semaphore for determining the status of the resource followed by the decision for proceeding.
- In multitasking operating systems, the activities are synchronized by using the semaphore techniques.
- Example, say we have four rooms with identical locks and keys. The semaphore count - the count of keys - is set to 4 at beginning (all four rooms are free), then the count value is decremented as people are coming in. If all rooms are full, ie. there are no free keys left, the semaphore count is 0. Now, when eq. one person leaves the rooms, semaphore is increased to 1 (one free key), and given to the next person in the queue.
- "A semaphore restricts the number of simultaneous users of a shared resource up to a maximum number. Threads can request access to the resource (decrementing the semaphore), and can signal that they have finished using the resource (incrementing the semaphore)."





Semaphore

- Synchronization tool that provides more sophisticated ways (than Mutex locks) for processes to synchronize their activities.
- Semaphore **S** – integer variable
- Can only be accessed via two indivisible (atomic) operations
 - **wait()** and **signal()**
 - ▶ Originally called **P()** and **V()**
- Definition of the **wait()** operation

```
wait(S) {  
    while (S <= 0)  
        ; // busy wait  
    S--;  
}
```

- Definition of the **signal()** operation

```
signal(S) {  
    S++;  
}
```

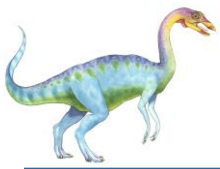




Semaphore (Cont.)

- **Counting semaphore** – integer value can range over an unrestricted domain
- **Binary semaphore** – integer value can range only between 0 and 1
 - Same as a **mutex lock**
- Can implement a counting semaphore **S** using binary semaphores
- With semaphores we can solve various synchronization problems





Semaphore Usage Example

- Solution to the CS Problem
 - Create a semaphore “**mutex**” initialized to 1
 - Code:

```
wait(mutex) ;
```

```
CS
```

```
signal(mutex) ;
```





Semaphore Usage Example (Cont.)

- Consider P_1 and P_2 that with two statements S_1 and S_2 and the requirement that S_1 to happen before S_2
 - Create a semaphore “**synch**” initialized to 0
 - Code:

P1:

$S_1;$

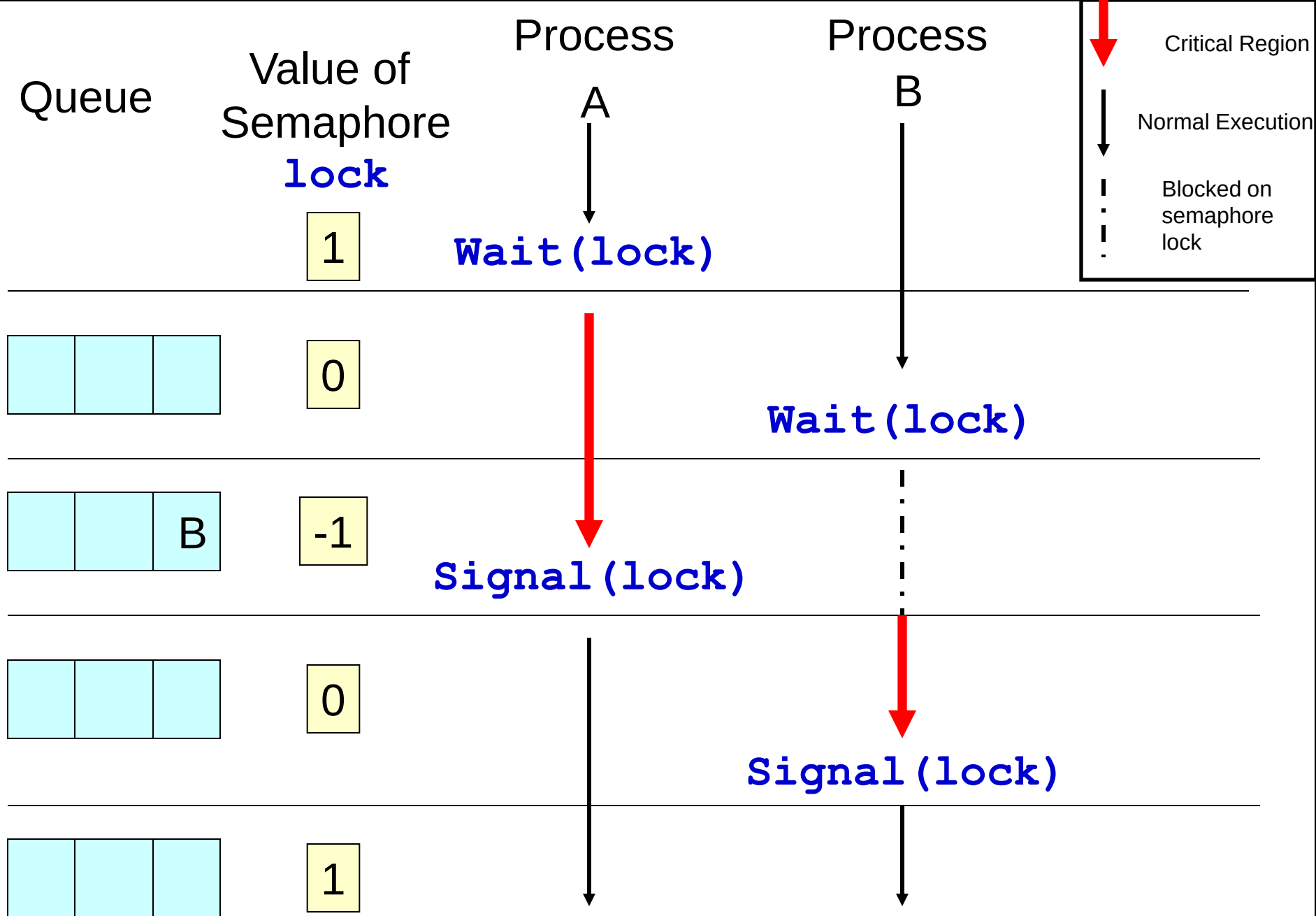
signal (synch) ;

P2:

wait (synch);

$S_2;$







Semaphore Implementation

- Must guarantee that no two processes can execute the **wait()** and **signal()** on the same semaphore at the same time
- Can be implemented using any of the critical sections solutions we discussed where the **wait** and **signal** code are placed in the critical section
- Could now have **busy waiting** in critical section implementation
 - But implementation code is short
 - Little busy waiting if critical section rarely occupied
- Note that for “regular” applications, where the application may spend lots of time in critical sections this is not a good solution





Implementation with no Busy waiting

- With each semaphore there is an associated waiting queue.
- Each entry in a waiting queue has two data items:
 - Value (of type integer)
 - Pointer to next record in the list
- Waiting queue

```
typedef struct {  
    int value;  
    struct process *list;  
} semaphore;
```





Implementation of the wait operation

- The **wait** operation:

```
wait(semaphore *S) {  
    S->value--;  
    if (S->value < 0) {  
        add this process to S->list;  
        sleep();  
    }  
}
```

- The **sleep()** suspends the process that invoked it.





Implementation of the signal operation

- The **signal** operation:

```
signal(semaphore *S) {  
    S->value++;  
    if (S->value <= 0) {  
        remove a process P from S->list;  
        wakeup(P);  
    }  
}
```

- The **wakeup(P)** operation resumes the execution of process P





Semaphore Example 1

```
semaphore s = 2;
```

```
Pi {
```

```
    while(1) {
```

```
        semWait(s);
```

```
        /* CS */
```

```
        semSignal(s);
```

```
        /* remainder */
```

```
    }
```

```
}
```

- What happens?
- When might this be desirable?





Semaphore Example 1

```
semaphore s = 2;
```

```
Pi {
```

```
    while(1) {
```

```
        semWait(s);
```

```
        /* CS */
```

```
        semSignal(s);
```

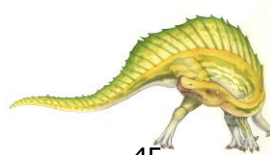
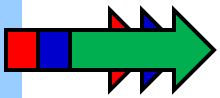
```
        /* remainder */
```

```
    }
```

```
}
```

$s = 2 \times \cancel{2} \times \cancel{0} \times -1$

- What happens?
- When might this be desirable?





Semaphore Example 1

```
semaphore s = 2;
```

```
Pi {  
    while(1) {  
        semWait(s);  
        /* CS */  
        semSignal(s);  
        /* remainder */  
    }  
}
```

- What happens?
 - Allows up to 2 processes to enter CS
- When might this be desirable?
 - Need up to 2 processes inside CS
 - ▶ e.g., limit number of processes reading a var
 - Be careful not to violate mutual exclusion inside CS!



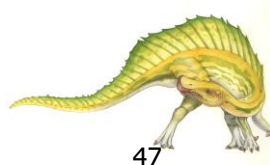


Semaphore Example 2

```
semaphore s = 0;
```

```
Pi {  
    while(1) {  
        semWait(s);  
        /* CS */  
        semSignal(s);  
        /* remainder */  
    }  
}
```

- What happens?
- When might this be desirable?





Semaphore Example 2

```
semaphore s = 0;
```

```
Pi {
```

```
    while(1) {
```

```
        semWait(s);
```

```
        /* CS */
```

```
        semSignal(s);
```

```
        /* remainder */
```

```
    }
```

```
}
```

■ What happens?

■ When might this be desirable?

$s = 0 \quad \times \quad -2 \quad \times \quad -2 \quad \times \quad -3$





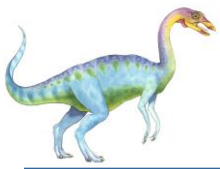
Semaphore Example 2

```
semaphore s = 0;
```

```
Pi {  
    while(1) {  
        semWait(s);  
        /* CS */  
        semSignal(s);  
        /* remainder */  
    }  
}
```

- What happens?
 - No one can enter CS! Ever!
- When might this be desirable?
 - Never!





Semaphore Example 3

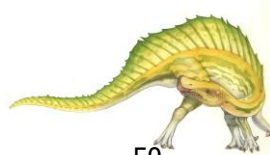
```
semaphore s = 0;
```

```
P1 {  
    /* do some stuff */  
    semWait(s);  
    /* do some more stuff */  
}
```

```
semaphore s; /* shared */
```

```
P2 {  
    /* do some stuff */  
    semSignal(s);  
    /* do some more stuff */  
}
```

- What happens?
- When might this be desirable?





Semaphore Example 3

```
semaphore s = 0;
```

```
P1 {
```

```
    /* do some stuff */
```



```
    semWait(s);
```

```
    /* do some more stuff */
```

```
}
```

- What happens?

```
semaphore s; /* shared */
```

```
P2 {
```

```
    /* do some stuff */
```



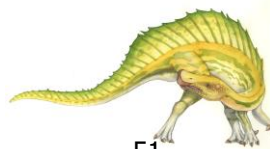
```
    semSignal(s);
```

```
    /* do some more stuff */
```

```
}
```

$s = \cancel{1} \cancel{0} 1$

- When might this be desirable?





Semaphore Example 3

```
semaphore s = 0;
```

```
P1 {
```

```
    /* do some stuff */
```

```
    semWait(s);
```

```
    /* do some more stuff */
```

```
}
```

```
semaphore s; /* shared */
```

```
P2 {
```

```
    /* do some stuff */
```

```
    semSignal(s);
```

```
    /* do some more stuff */
```

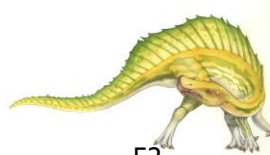
```
}
```

■ What happens?

- P1 waits until P2 signals
- if P2 signals first, P1 does not wait

■ When might this be desirable?

- Having a process/thread wait for another process/thread





GATE 2016 QUESTION

- (1) Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____. (A) 7 (B) 8 (C) 9 (D) 10
- (2) A counting semaphore was initialized to 10. Then 6 P (wait) operations and 4 V (signal) operations were completed on this semaphore. The resulting value of the semaphore is
- (3) Let S be the binary semaphore variable $S = 0$ initially. Assume that no blocked processes exist in the system. The following signal (V), wait (P) operations are performed. The number of blocked processes at the end are _____.

4 P, 6 V, 9 P, 13 V, 14 P





Problems with Semaphores

- Incorrect use of semaphore operations:
 - `signal(mutex) ... wait(mutex)`
 - `wait(mutex) ... wait(mutex)`
 - Omitting of `wait(mutex)` and/or `signal(mutex)`
- These – and others – are examples of what can occur when semaphores and other synchronization tools are used incorrectly.
- Solution: introduce high-level programming constructs





Monitors

- A high-level abstraction that provides a convenient and effective mechanism for process synchronization
- *Abstract data type*, internal variables only accessible by code within the procedure
- Only one process may be active within the monitor at a time
- Pseudocode syntax of a monitor:

```
monitor monitor-name
{
    // shared variable declarations
    function P1 (...) { ... }

    function P2 (...) { ... }

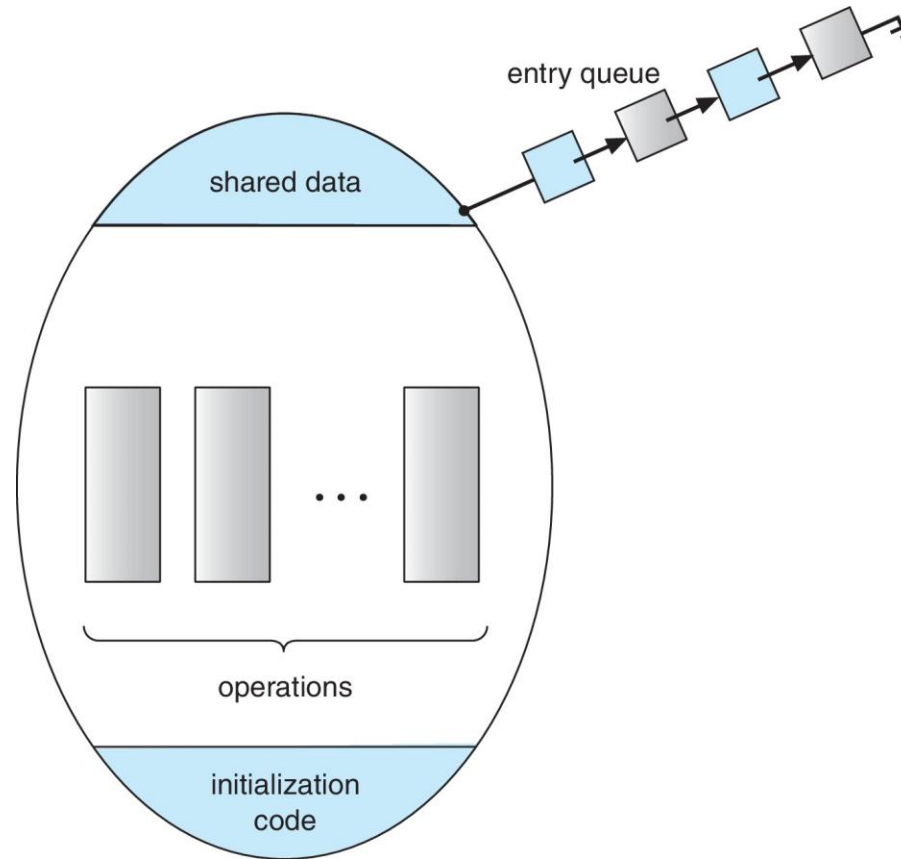
    function Pn (...) {.....}

    initialization code (...) { ... }
}
```





Schematic view of a Monitor





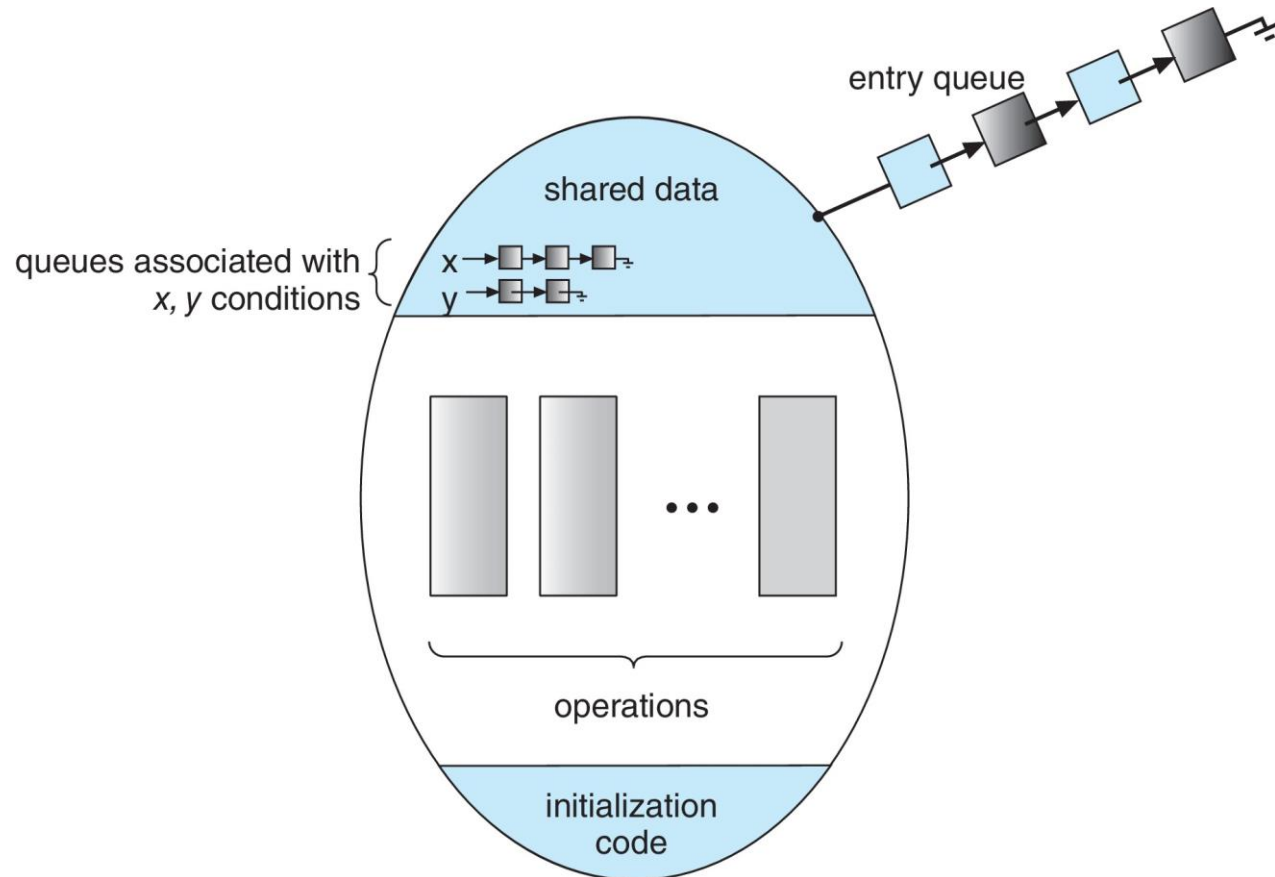
Condition Variables

- **condition x , y ;**
- Two operations are allowed on a condition variable:
 - **$x.\text{wait}()$** – a process that invokes the operation is suspended until **$x.\text{signal}()$**
 - **$x.\text{signal}()$** – resumes one of processes (if any) that invoked **$x.\text{wait}()$**
 - ▶ If no **$x.\text{wait}()$** on the variable, then it has no effect on the variable





Monitor with Condition Variables





Condition Variables Choices

- If process P invokes **`x.signal()`** , and process Q is suspended in **`x.wait()`** , what should happen next?
 - Both Q and P cannot execute in parallel. If Q is resumed, then P must wait
- Options include
 - **Signal and wait** – P waits until Q either leaves the monitor or it waits for another condition
 - **Signal and continue** – Q waits until P either leaves the monitor or it waits for another condition
 - Both have pros and cons – language implementer can decide
 - Monitors implemented in Concurrent Pascal compromise
 - ▶ P executing signal immediately leaves the monitor, Q is resumed
 - Implemented in other languages including Mesa, C#, Java





Monitor Implementation Using Semaphores

- Variables

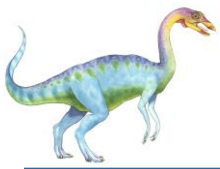
```
semaphore mutex;  // (initially = 1)
semaphore next;   // (initially = 0)
int next_count = 0; // number of processes waiting
                    inside the monitor
```

- Each function F will be replaced by

```
wait(mutex);
...
body of F;
...
if (next_count > 0)
    signal(next)
else
    signal(mutex);
```

- Mutual exclusion within a monitor is ensured





Implementation – Condition Variables

- For each condition variable x , we have:

```
semaphore x_sem; // (initially = 0)
int x_count = 0;
```

- The operation $x.\text{wait}()$ can be implemented as:

```
x_count++;
if (next_count > 0)
    signal(next);
else
    signal(mutex);
wait(x_sem);
x_count--;
```





Implementation (Cont.)

- The operation `x.signal()` can be implemented as:

```
if (x_count > 0) {  
    next_count++;  
    signal(x_sem);  
    wait(next);  
    next_count--;  
}
```





Resuming Processes within a Monitor

- If several processes are queued on condition variable **x**, and **x.signal()** is executed, which process should be resumed?
- FCFS frequently not adequate
- **conditional-wait** construct of the form **x.wait(c)**
 - Where **c** is **priority number**
 - Process with lowest number (highest priority) is scheduled next





Single Resource allocation Example

- Allocate a single resource among competing processes using priority numbers that specify the maximum amount of time a process plans to use the resource

R.acquire(t) ;

...

access the resource;

...

R.release;

- Where R is an instance of type **ResourceAllocator** (shown in the next slides)

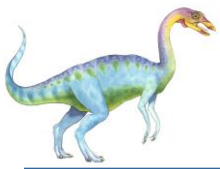




A Monitor to Allocate Single Resource

```
monitor ResourceAllocator
{
    boolean busy;
    condition x;
    void acquire(int time) {
        if (busy)
            x.wait(time);
        busy = true;
    }
    void release() {
        busy = FALSE;
        x.signal();
    }
    initialization code() {
        busy = false;
    }
}
```





Single Resource Monitor (Cont.)

- Usage:
 acquire
 ...
 release
- Incorrect use of monitor operations
 - **release()** ... **acquire()**
 - **acquire()** ... **acquire()**
 - Omitting of **acquire()** and/or **release()**





Liveness

- Processes may have to wait indefinitely while trying to acquire a synchronization tool such as a mutex lock or semaphore.
- Waiting indefinitely violates the progress and bounded-waiting criteria discussed at the beginning of this chapter.
- **Liveness** refers to a set of properties that a system must satisfy to ensure processes make progress.
- Indefinite waiting is an example of a liveness failure.





Deadlock

- **Deadlock** – two or more processes are waiting indefinitely for an event that can be caused by only one of the waiting processes
- Let S and Q be two semaphores initialized to 1

P_0	P_1
<code>wait(S) ;</code>	<code>wait(Q) ;</code>
<code>wait(Q) ;</code>	<code>wait(S) ;</code>
<code>...</code>	<code>...</code>
<code>signal(S) ;</code>	<code>signal(Q) ;</code>
<code>signal(Q) ;</code>	<code>signal(S) ;</code>

- Consider if P_0 executes `wait(S)` and P_1 `wait(Q)`. When P_0 executes `wait(Q)`, it must wait until P_1 executes `signal(Q)`
- However, P_1 is waiting until P_0 execute `signal(S)`.
- Since these `signal()` operations will never be executed, P_0 and P_1 are **deadlocked**.



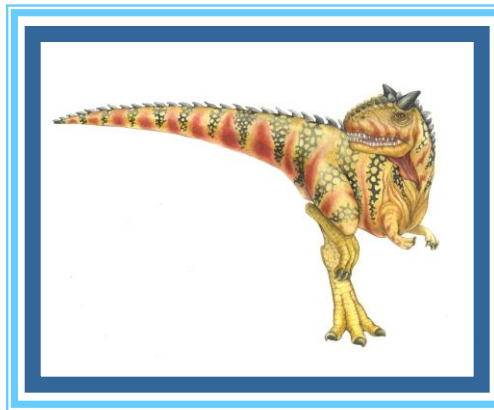


Other Forms of Deadlock

- **Starvation** – indefinite blocking
 - A process may never be removed from the semaphore queue in which it is suspended
- **Priority Inversion** – Scheduling problem when lower-priority process holds a lock needed by higher-priority process
- Solved via **priority-inheritance protocol**



End of Chapter 6





Priority Inheritance Protocol

- Consider the scenario with three processes **P1**, **P2**, and **P3**. **P1** has the highest priority, **P2** the next highest, and **P3** the lowest.
- Assume a resource **P3** is assigned a resource **R** that **P1** wants.
 - Thus, **P1** must wait for **P3** to finish using the resource. However, **P2** becomes runnable and preempts **P3**.
 - What has happened is that **P2** - a process with a lower priority than **P1** - has indirectly prevented **P3** from gaining access to the resource.
- To prevent this from occurring, a **priority inheritance protocol** is used. This simply allows the priority of the highest thread waiting to access a shared resource to be assigned to the thread currently using the resource. Thus, the current owner of the resource is assigned the priority of the highest priority thread wishing to acquire the resource.





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